

$$a) \quad Y(n) = X(n-3) + X(n-2) + X(n-1) + X(n)$$

$$Y(z) = X(z) (z^{-3} + z^{-2} + z^{-1} + 1)$$

$$\frac{Y(z)}{X(z)} = z^{-3} + z^{-2} + z^{-1} + 1 \Rightarrow T(z) = 1 + z^{-1} + z^{-2} + z^{-3}$$

$$\begin{aligned} T(e^{i\Omega}) &= 1 + e^{-i\Omega} + e^{-2i\Omega} + e^{-3i\Omega} = e^{-i\frac{3}{2}\Omega} (e^{i\frac{3}{2}\Omega} + e^{i\frac{1}{2}\Omega} + e^{-i\frac{1}{2}\Omega} + e^{-i\frac{3}{2}\Omega}) \\ &= 2e^{-i\frac{3}{2}\Omega} [\cos(\frac{3}{2}\Omega) + \cos(\frac{1}{2}\Omega)] \quad \left\{ \begin{array}{l} |T(e^{i\Omega})| = 2 |\cos(\frac{3}{2}\Omega) + \cos(\frac{1}{2}\Omega)| \\ \angle T(e^{i\Omega}) = -\frac{3}{2}\Omega \end{array} \right. \end{aligned}$$

$$b) \quad Y(n) = X(n-4) + X(n-3) + X(n-2) + X(n-1) + X(n)$$

$$Y(z) = X(z) (z^{-4} + z^{-3} + z^{-2} + z^{-1} + 1)$$

$$T(z) = \frac{Y(z)}{X(z)} = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

$$\begin{aligned} T(e^{i\Omega}) &= 1 + e^{-i\Omega} + e^{-2i\Omega} + e^{-3i\Omega} + e^{-4i\Omega} \\ &= e^{-2i\Omega} (e^{2i\Omega} + e^{i\Omega} + 1 + e^{-i\Omega} + e^{-2i\Omega}) \\ &= e^{-2i\Omega} [1 + 2\cos(\Omega) + 2\cos(2\Omega)] \quad \left\{ \begin{array}{l} |T(e^{i\Omega})| = |1 + 2\cos(\Omega) + 2\cos(2\Omega)| \\ \angle T(e^{i\Omega}) = -2\Omega \end{array} \right. \end{aligned}$$

$$c) \quad Y(n) = X(n) - X(n-1)$$

$$Y(z) = X(z) (1 - z^{-1})$$

$$\begin{aligned} T(z) = \frac{Y(z)}{X(z)} &= 1 - z^{-1} \Rightarrow T(e^{i\Omega}) = 1 - e^{-i\Omega} = e^{-i\frac{\Omega}{2}} (e^{i\frac{\Omega}{2}} - e^{-i\frac{\Omega}{2}}) \\ &= 2i e^{-i\frac{\Omega}{2}} \sin(\frac{\Omega}{2}) \quad \left\{ \begin{array}{l} |T(e^{i\Omega})| = 2 |\sin(\frac{\Omega}{2})| \\ \angle T(e^{i\Omega}) = \frac{\pi}{2} - \frac{\Omega}{2} \end{array} \right. \end{aligned}$$

$$d) \quad Y(n) = X(n) - X(n-2)$$

$$Y(z) = X(z) (1 - z^{-2})$$

$$T(z) = \frac{Y(z)}{X(z)} = 1 - z^{-2}$$

$$\begin{aligned} T(e^{i\Omega}) &= 1 - e^{-2i\Omega} = e^{-i\Omega} (e^{i\Omega} - e^{-i\Omega}) = 2i e^{-i\Omega} \sin(\Omega) \quad \left\{ \begin{array}{l} |T(e^{i\Omega})| = 2 |\sin(\Omega)| \\ \angle T(e^{i\Omega}) = \frac{\pi}{2} - \Omega \end{array} \right. \end{aligned}$$